

Monitoring global power outages induced by tropical cyclones using nighttime light data

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ABSTRACT

Tropical cyclones (TCs) are among the most destructive natural hazards, frequently causing widespread power outages (POs) in coastal urban areas that disrupt economic activity and social stability. Quantifying TC-induced POs remains challenging due to limited outage data availability. This study made the first global detection and quantification of TC-induced POs using NASA's Black Marble nighttime lights (NTL) data. The proposed method analyzed changes in NTL brightness within urban agglomerations by establishing pre-TC baselines and applying statistical outlier detection to identify outages. A total of 1,239 POs were detected by the algorithm from 19,999 agglomeration-TC events between 2012 and 2023, with the corresponding outage duration and severity being also estimated. Validation against media reports showed an overall accuracy of 0.78, with the accuracy being improved with TC intensity. Case studies demonstrated robust performance in regions with vulnerable infrastructure and high-quality NTL observations, such as North America, while performance declined in areas affected by frequent data gaps or rapid restoration, notably East Asia and India. While only 50% of agglomeration-TC events can be evaluated due to the missing NTL data, this work offers a scalable, near-real-time approach to global TC-induced PO monitoring, providing critical insights for urban resilience planning, disaster response, and power system management.

1. Introduction

Climate change is contributing to a global increase in the frequency and intensity of extreme weather events (Stott, 2016; Stott et al., 2016). Among these, Tropical Cyclones (TCs) are among the most destructive, frequently causing large-scale Power Outages (POs) in coastal regions and posing severe threats to economic development and social stability. For example, TC Sandy in 2012 resulted in POs affecting over 8 million residents, with economic losses reaching ~\$65 billion (Zamuda et al., 2013). Similarly, TC Maria in 2017 devastated Puerto Rico's power grid, resulting in the most severe PO in U.S. history that last over ten months (Griffin and Johnston, 2017; Kwasinski et al., 2019). In eastern Asia, TCs such as Damrey in 2005, Hagupit in 2008, Ute and Usagi in 2013, and Rammasun in 2014 repeatedly caused transmission failures and widespread grid disruptions (Li et al., 2015a; Tang et al., 2006).

Power infrastructure is a critical foundation of modern society, and

outages can trigger cascading effects, such as disruptions in food and water supply, communication breakdowns, increased crime rates, and inaccessibility of medical equipment (Hasan and Foliente, 2015; Zimmerman et al., 2017). In a warming world, the likelihood of major TCs is steadily increasing (Knutson et al., 2020), particularly in the North Atlantic, where their occurrence is projected to rise by 49% per decade (Kossin et al., 2020). In the continental United States, both the average and maximum rainfall associated with TCs have been increasing (Uehling and Schreck III, 2024). Furthermore, the destructive impact of landfalling TCs in China has also shown a significant upward trend (Liu et al., 2020). In this context, quantifying the impact of TCs on power infrastructure is crucial. Effective quantification methods not only deepen the understanding of extreme weather effects on power systems but also offer valuable insights for developing mitigation strategies and enhancing grid resilience.

Various approaches have been developed to assess the impact of TCs

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on urban power systems. Statistical models, such as generalized linear models (Liu et al., 2008; Liu et al., 2005) and generalized additive models (Han et al., 2009), have been employed to estimate TC-induced POs. More recently, machine learning techniques, e.g., random forest, have been applied to predict POs by integrating TC characteristics, distribution network damage reports, and local meteorological conditions (Nateghi et al., 2014; Oh et al., 2022; Yuan et al., 2020). Additionally, PO estimation models have been used in turn to simulate the effects of climate change on TC intensity, frequency, and trajectory, modeling their potential impact on power systems under various scenarios. While these studies have significantly advanced our understanding of power grid vulnerability and outage assessment in response to TCs, they primarily relied on outage data provided by local utility companies and public service departments. A major challenge in quantifying urban POs is the difficulty in obtaining comprehensive outage data. These records are often scattered across different years and regions, and in many cases, they are incomplete or entirely unavailable.

To address this issue, Remote sensing Nighttime light (NTL) data has emerged as a viable alternative due to its large-scale coverage, near real-time availability, and low cost (Cao et al., 2013). NTL data has been widely used in urbanization modeling (Chen et al., 2019; Zhou et al., 2018), socioeconomic analysis (e.g., population, GDP, Lo, 2001; Zhao et al., 2017), environmental monitoring (e.g., CO₂ emissions, Li et al., 2024), and the study of armed conflicts (Li et al., 2013; Li et al., 2015b) and natural disasters (Cao et al., 2013; Molthan and Jedlovec, 2013). Among the commonly used NTL products, NASA's Black Marble NTL product which is derived from the Suomi National Polar-orbiting Partnership Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band provides globally corrected daily NTL data (Wang et al., 2024). A few studies have applied NTL data in quantifying abrupt POs and restoration processes based on the abrupt change between pre- and post-cyclone NTL intensities (Azad and Ghandehari, 2021; Román et al., 2019; Wang et al., 2018; Xu et al., 2023). Key parameters such as initial restoration time, total restoration duration, and restoration rate can also be estimated from NTL data to monitor recovery process (Wang et al., 2018). More recently, Zheng et al. (2025) employed NTL time series to reveal substantial spatial heterogeneity and inequality in post-hurricane recovery, highlighting pronounced differences in recovery trajectories across regions and communities. These studies demonstrate the effectiveness of NTL data in assessing power loss spatially and temporally. However, current studies remain largely confined to case studies of major TC-induced PO events. Variations in socioeconomic conditions, infrastructure resilience, and grid structures across different regions pose challenges to the adaptability and generalization of existing approaches, requiring a method for conducting global-scale TC-induced PO. Moreover, a comprehensive evaluation scheme is required to determine how much the end users can trust the NTL-based results.

In this study, a global detection and quantification framework for TC-induced POs was developed using NASA's Black Marble nighttime light (NTL) data. The algorithm operated at the urban agglomeration scale, establishing a pre-cyclone NTL baseline and detecting POs as statistical outliers based on significant declines in post-TC NTL brightness. Both the severity and duration of outages are quantified by analyzing temporal changes in NTL intensity. The method was applied globally for the period 2012–2023, resulting in a comprehensive dataset of TC-induced PO events. Validation is conducted using media reports and documented case studies to assess detection accuracy. Furthermore, the spatial and temporal characteristics of outage severity and duration are examined to identify regional patterns and infrastructure vulnerabilities.

2. Data

2.1. NTL data

Remote sensing of night light starts from the Defense Meteorological Satellite Program (DMSP) in 1970 s, initially designed for monitoring

could at night. Early NTL products from DMSP Operational Linescan System (OLS) and its successive VIIRS sensors suffered from saturation in urban centers, the absence of onboard calibration, and contamination from atmospheric and lunar illumination (Cao et al., 2019; Letu et al., 2010; Levin et al., 2020; Ma et al., 2014). Accordingly, the daily moonlight-adjusted NTL product (VNP46A2/VJ146A2) and the annual moonlight-adjusted NTL product (VNP46A4/VJ146A4) from the Black Marble suite were used in this study to detect POs. These products were developed by NASA, offering global NTL observations at a 500-meter spatial resolution from January 2012 to the present, with an example shown in Fig. 1a. This enhanced product incorporates correction for atmospheric effects, terrain distortions, the lunar bidirectional reflectance distribution function, thermal effects, and stray light contamination (Wang et al., 2024). The saturation effect and onboard calibration issues of earlier products were also addressed, allowing minimal pre-processing for end users.

The daily NTL data used in this study was derived from the DNB_BRDF_Corrected_NTL product within the VNP46A2 dataset (Wang et al., 2024). Only areas labeled as “Persistent nighttime lights” in the Mandatory_Quality_Flag layer were used, resulting in high-quality, stable NTL values. Persistent NTL signals typically originate from consistent artificial light sources such as towns, roads, and industrial facilities, and exclude transient events like wildfires or temporary lighting. This focus on stable light sources enables a more accurate reflection of long-term economic activity and infrastructure, making the dataset well-suited for assessing power impacts caused by TCs. The zenith angle layer of the at-sensor top of atmosphere nighttime radiance data of the Black Marble Product suite (VNP46A1) was obtained for angular effect correction.

The annual NTL data was sourced from the AllAngle_Composite_Snow_Free product within the VNP46A4 dataset. This product provides an annual composite of snow-free, all-angle daily NTL radiance values. Similarly, only pixels labeled as “Good-quality” in the AllAngle_Composite_Snow_Free_Quality layer were included. The extracted dataset captures the average state of nighttime illumination over the year and was used as a robust reference for calibrating daily NTL observations.

2.2. Global TC records

TC records were obtained from the International Best Track Archive for Climate Stewardship (IBTrACS), which consolidates both the recent and historical TC data from multiple agencies into a centralized global best-track dataset (Knapp et al., 2010). Recognized by the World Meteorological Organization's Tropical Cyclone Programme, IBTrACS is currently the most comprehensive global TC dataset, designed to support improved understanding of TC distribution, frequency, and intensity. The archive provides detailed information for each cyclone from 1992 to the present, including identification codes, names, timestamps, geographic tracks, central wind speeds, atmospheric pressure, and wind radii. It was served as the primary data source for TC characterization in this study. Based on China's 2006 classification scheme, the intensity levels and tracks of global TCs from 2012 to 2023 are illustrated in Fig. 1a.

2.3. Landcover data

The global land cover map series from the European Space Agency (ESA) Climate Change Initiative Land Cover (CCI_LC, 1992–2015, Herold et al., 2006) and the Copernicus Climate Change Service Land Cover (C3S_LC, 2016–2021, Defourny et al., 2021) was used to identify urban agglomerations. The C3S_LC dataset is a continuation of the CCI_LC data, together providing global annual land cover maps at a spatial resolution of 300 m from 1992 to 2021. To ensure consistency with NTL data, these maps were resampled to a resolution of 500 m using the nearest neighbor sampling method. Each pixel in the land cover map represents

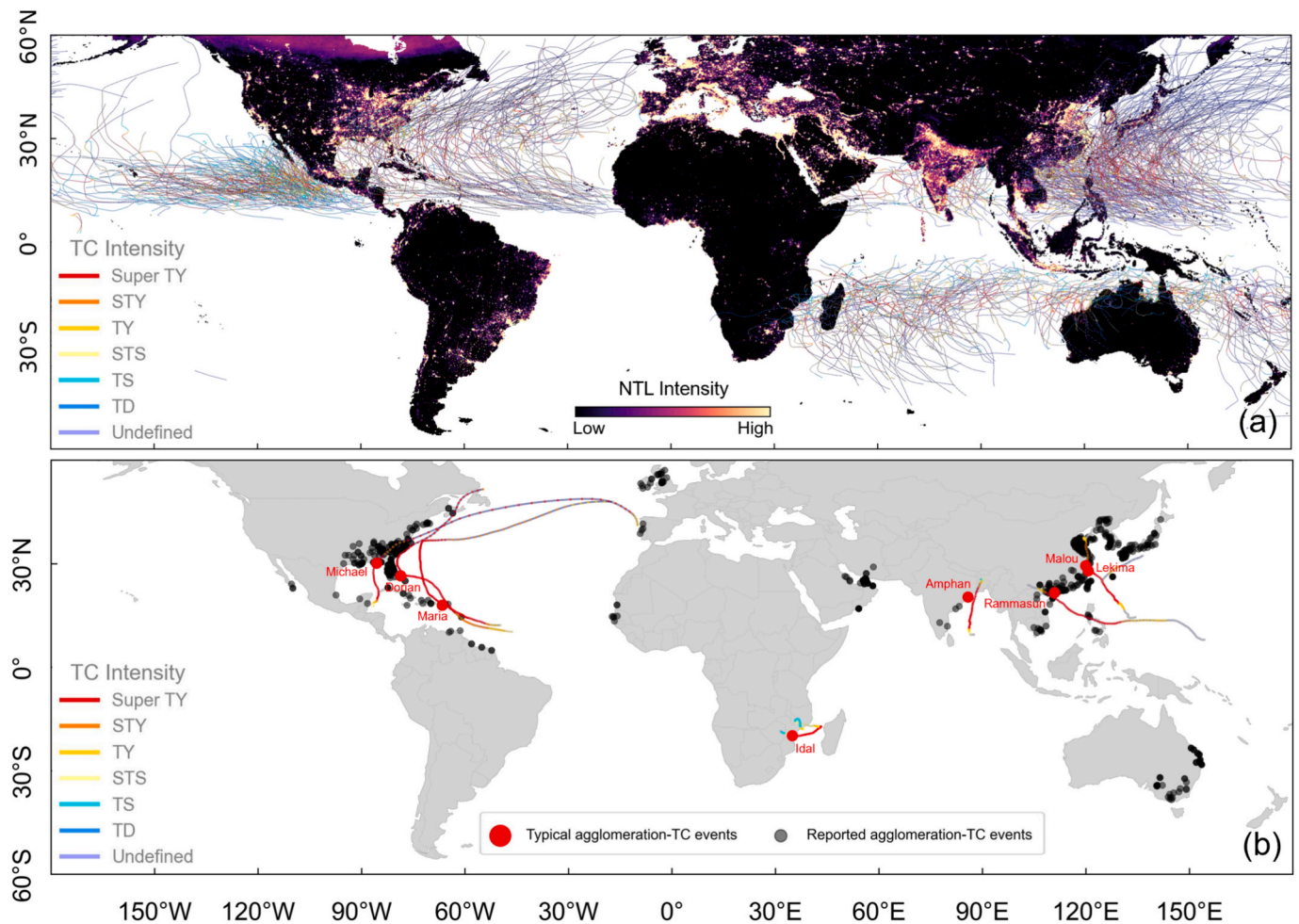


Fig. 1. (a) Annual NASA Black Marble Nighttime Light (NTL) data for 2023 and Tropical Cyclone (TC) intensity and tracks from 2012 to 2023, classified based on China's 2006 standard. Categories include Tropical Depression (TD, ≤ 33.3 kt), Tropical Storm (TS, 33.3–47.4 kt), Severe Tropical Storm (STS, 47.6–63.4 kt), Typhoon (TY, 63.6–80.5 kt), Severe Typhoon (STY, 80.6–98.9 kt), and Super Typhoon (Super TY, ≥ 99.1 kt). TC tracks data were obtained from the International Best Track Archive for Climate Stewardship. (b) Spatial distribution of 721 urban agglomeration-TC impact events validated through Google search results. Eight representative events selected for detailed analysis are highlighted in red. Corresponding cyclone tracks and intensity levels are also shown.

a land cover class based on the United Nation Land Cover Classification System, with urban areas assigned a fixed value of 190. The urban pixels were then extracted to generate annual urban mask data.

2.4. Media records for validation

Online media reports were used as external validation sources for the algorithm. A total of 88 TCs were randomly selected, and information on the presence or absence of POs was retrieved through Google search results. These TCs affected 721 city-level locations, defined here as instances where a city experienced TC impact. Among them, 249 PO events were identified (Fig. 1b), with 226 containing specific records of the TC impact onset. The corresponding references are listed in the [supplementary materials](#). In addition, eight representative events from different regions were selected for detailed case analysis (Fig. 1b).

As most media reports are provided at the provincial or state level and rarely offer information at the urban agglomeration scale, the determination of PO occurrence was based on whether the city was within the affected area of TC-induced POs. Additionally, except in cases of major POs, the reported TC impact onset typically refers to the province or state in which the impacted city is located. Given that TCs generally traverse a province or state within approximately one day (Yamaguchi et al., 2020), the reported TC impact onset may differ from the actual onset time of local impacts by about one day.

3. Methods

The proposed algorithm consists of two main components: extraction of NTL time series for urban agglomerations and identification of TC-induced POs (Fig. 2). Daily averaged NTL time series were first extracted for each urban agglomeration by integrating land cover data, NTL observations, and TC records. These time series were then used to detect TC-induced POs and to quantify their associated duration and severity.

3.1. Extraction of daily averaged NTL for urban agglomerations

Daily NTL images remain subject to an angular effect, leading to inconsistencies among observations of the same location across different Viewing Zenith Angles (VZA) (Li et al., 2022; Tan et al., 2022). Following previous studies (Hu et al., 2024; Xu et al., 2023), a quadratic regression approach consistent with that employed by Zheng et al. (2025) was applied to normalize NTL radiance observed at different VSAs to the radiance at nadir ($VZA = 0^\circ$). The angular effect was found to contribute to an NTL radiance bias within 2.5% of nadir observations (Fig. S1 in [supplementary material](#)).

This study focuses on urban agglomerations as the spatial analysis units, a choice that helps mitigate the impact of missing pixel data due to cloud cover and enhances the completeness and analytical robustness of NTL observations. Urban agglomerations were delineated using urban

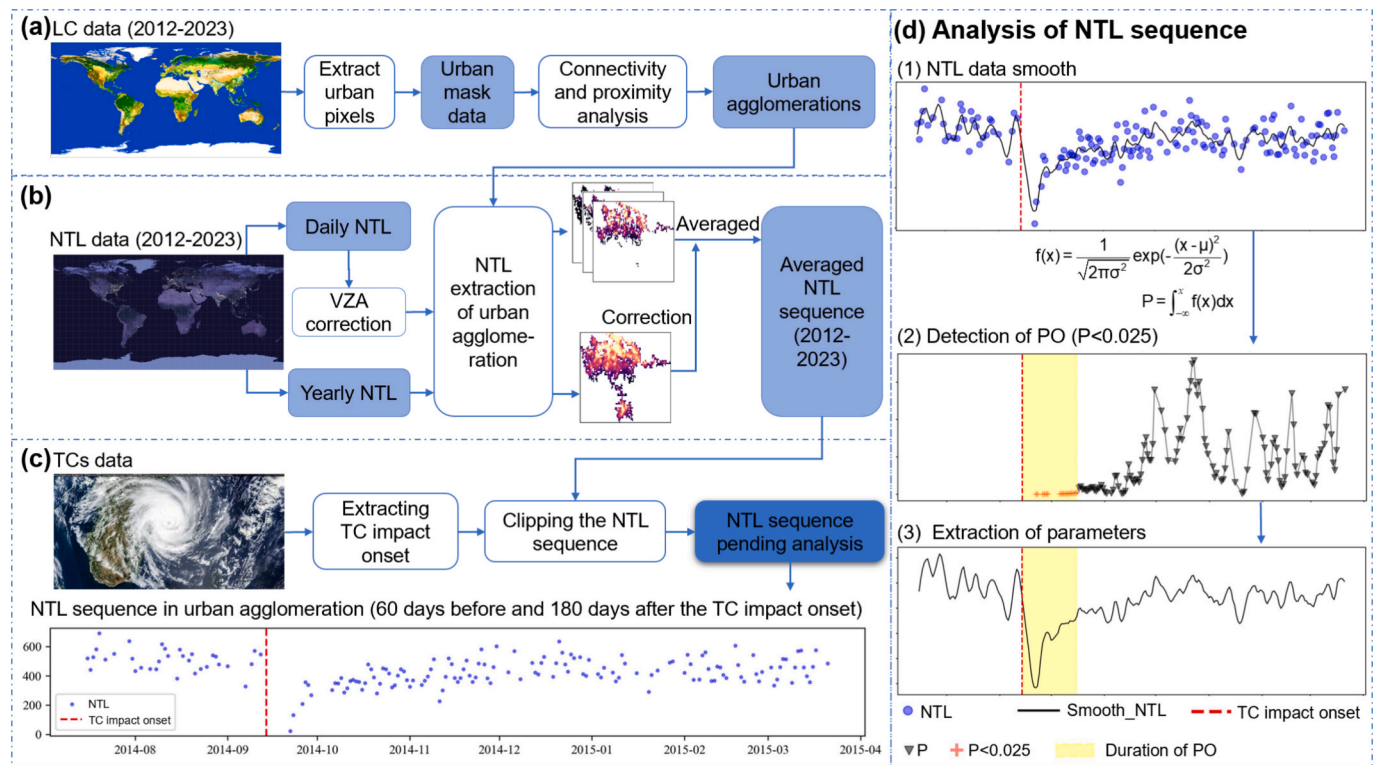


Fig. 2. The proposed Power Outage (PO) detection method. The framework integrates land cover, NTL, and TC data to extract daily averaged NTLs over urban agglomerations and applies an outlier-based algorithm to identify and quantify TC-induced POs.

mask data derived from the C3S_LC and CCI_LC datasets. As land cover data are only available up to 2021, and urban expansion over short timeframes is generally limited, the 2021 data were used to represent urban extent in 2022 and 2023 to ensure temporal consistency. Urban agglomerations in this study were defined as contiguous areas exceeding 10×10 pixels (approximately 25 km^2). Additionally, adjacent urban patches located within 10 km of an agglomeration were merged into a single entity using a graph-based clustering method, capturing the spatial continuity of real-world urban conurbations. This approach aligns with the functional dynamics of urban grid networks and reduces sensitivity to minor classification errors in land cover data by emphasizing larger, connected urban clusters.

Daily averaged NTL values were calculated for each urban agglomeration. However, frequent cloud cover often limits the availability of valid NTL observations, which may fail to fully capture the spatial extent

of an agglomeration (as illustrated in Fig. 3), resulting in temporal variability in the proportion of observed pixels. Directly averaging the available NTL values may therefore lead to biased estimates and introduce random fluctuations in the time series. To address this issue, it was assumed that the ratio of NTL brightness between observed (valid) and unobserved (missing) pixels in the daily dataset is consistent with that in the corresponding annual composite. Under this assumption, the daily NTL value of missing pixels $NTL_{\text{daily},M}$ can be estimated by:

$$NTL_{\text{daily},M} = \frac{NTL_{\text{annual},M}}{NTL_{\text{annual},V}} NTL_{\text{daily},V} \quad (1)$$

where M and V represent the masks of missing pixels and valid pixels in the daily NTL image respectively. This approach relies on the additional assumption that the spatial distribution of light emissions in daily observations does not deviate substantially from the annual average, and that the effects of TC-induced outages are relatively uniform across the agglomeration.

Lighting patterns may evolve rapidly during the recovery process. To evaluate the robustness of the assumption under dynamic conditions, a temporal sensitivity analysis was made on three representative large-scale PO events (Fig. 1) with restoration periods exceeding 30 days. For each event, three distinct temporal windows were defined, including the pre-outage (two months prior to the event), outage (from the onset of the event to full restoration), and post-outage (two months following full recovery). To simulate the impact of cloud cover during these phases, synthetic missing data patches were generated. For every day with available observations, a random pixel within the affected urban agglomeration was selected as a seed point, followed by spatial dilation to create a missing region covering 30–70% of the total area (at 10% increments). The NTL brightness ratios between the daily observations and the annual composites within these synthetic patches were compared (e.g., Eq. (1)).

The results indicate high consistency during the stable pre- and post-outage periods, with a Root Mean Square Difference (RMSD) of 2–4%

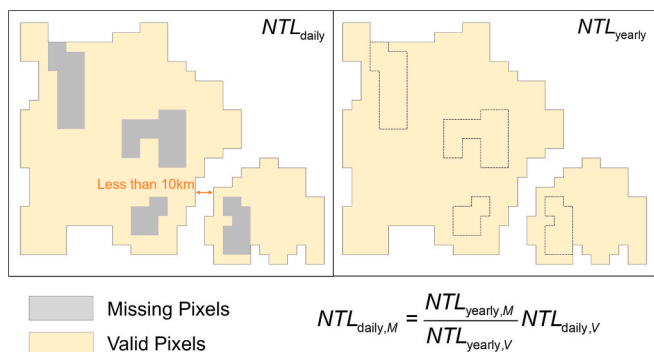


Fig. 3. Schematic illustration showing daily NTL data affected by cloud cover and the corresponding annual averaged NTL image for a selected urban agglomeration. Valid and missing pixels are indicated to demonstrate how daily NTL gaps are filled using annual reference data. Two urban patches with a distance of $< 10 \text{ km}$ were treated as the same agglomeration.

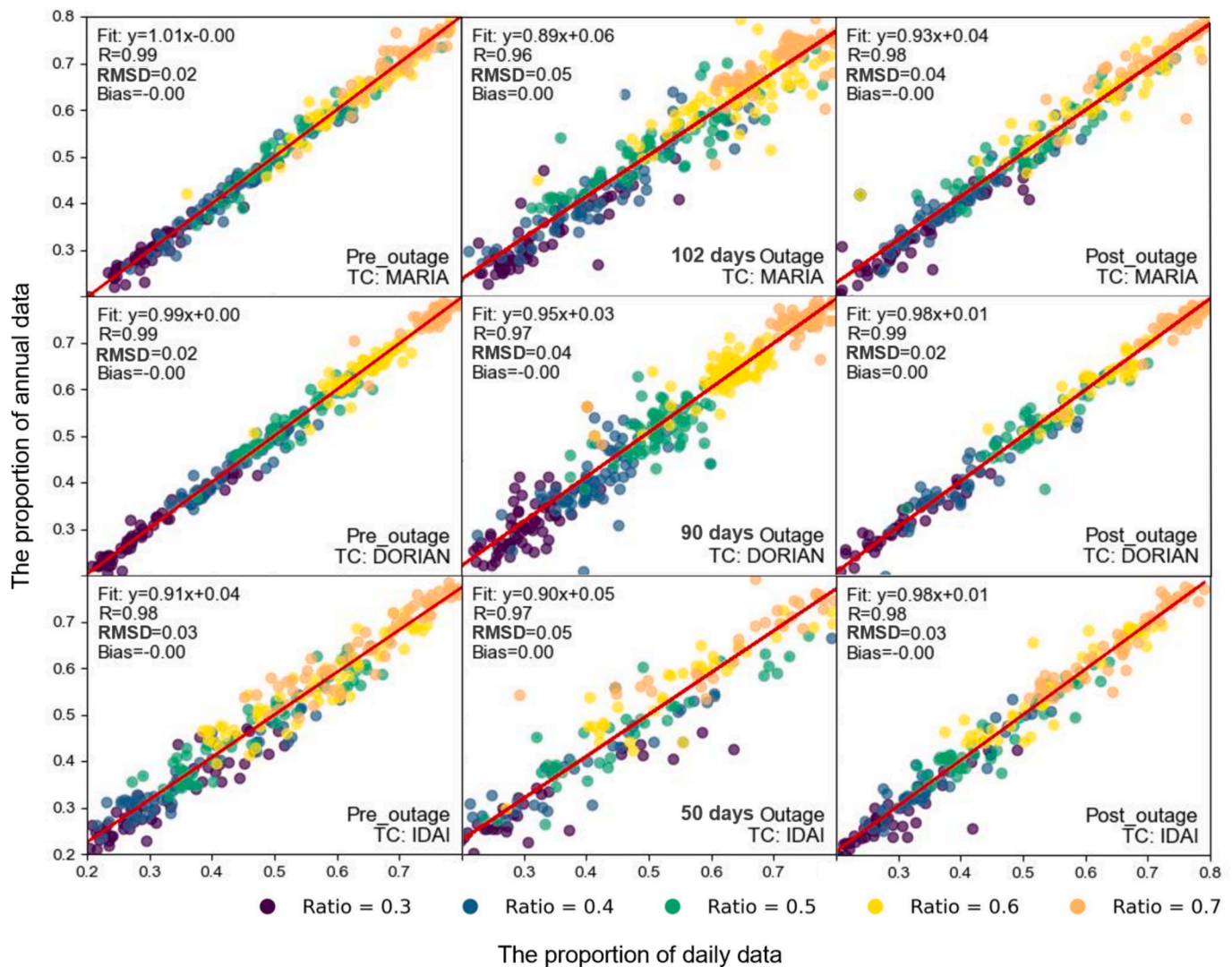


Fig. 4. Sensitivity analysis of the spatial consistency assumption (Eq. (1)) across different disaster phases. The scatter plots compare the NTL brightness ratio of synthetic missing patches relative to the total urban agglomeration at the daily scale versus the corresponding ratio in the annual composite. Results are shown for three representative large-scale power PO events across three temporal windows: Pre-outage (baseline stability), Outage (dynamic recovery phase), and Post-outage (restored stability). Missing data proportions were simulated from 0.3 to 0.7 at 0.1 increments.

(Fig. 4). During the dynamic outage phase, the RMSD increased slightly to 4%–5%, reflecting the expected spatial heterogeneity of infrastructure restoration. However, this level of uncertainty is substantially lower than the magnitude of temporal anomalies caused by the disasters themselves as well as the random fluctuations in the original time series (Fig. 5). Since the primary focus of this study is on recovery patterns at the agglomeration scale rather than intra-urban variability, the assumption in Eq. (1) remains a robust proxy for reconstructing missing daily NTL data.

For missing pixel issues, seasonal variations in nighttime light intensity can introduce noise into time series analyses (Levin, 2017). Although such variations are typically minor in urban agglomerations, they were removed by differencing the NTL values from the same period in the nearest previous year in which no abrupt changes were detected.

3.2. Smoothing the daily NTL sequences of urban agglomerations

To reduce high-frequency fluctuations and emphasize underlying trends in the daily NTL time series, the Savitzky–Golay (S-G) filter was applied to each urban agglomeration's NTL sequence. The S-G filter is widely used for time-series smoothing and denoising in remote sensing

applications (Chen et al., 2021; Jiao and Wang, 2021; Wen et al., 2023). It operates by fitting successive subsets of adjacent data points with a low-degree polynomial using the least-squares method, allowing for noise suppression while preserving the essential shape and features of the original signal (Savitzky and Golay, 1964).

The performance of the S-G filter depends primarily on two parameters: the window size and the polynomial order. Larger window sizes improve noise suppression, while higher polynomial orders help retain signal details (Luo et al., 2005). Multiple combinations of window sizes (5, 9, 13, and 17) and polynomial orders (1, 2, and 3) were tested to determine the empirically optimal combination. The effectiveness of each combination was assessed by comparing the smoothed NTL sequences before and after TC impact onset across 20 selected urban agglomerations (Fig. S2). Results for two representative agglomerations are presented in Fig. 5. A window size of 9 and a polynomial order of 2 was found to yield the best balance between noise reduction and the preservation of post-impact NTL minima. This configuration smooths the signal across a 9-day window, which introduces a temporal uncertainty of approximately ± 4 –5 days. However, sensitivity tests indicated that variations in filter settings had only a marginal effect on the detection of power outages. Additional results and comparisons are

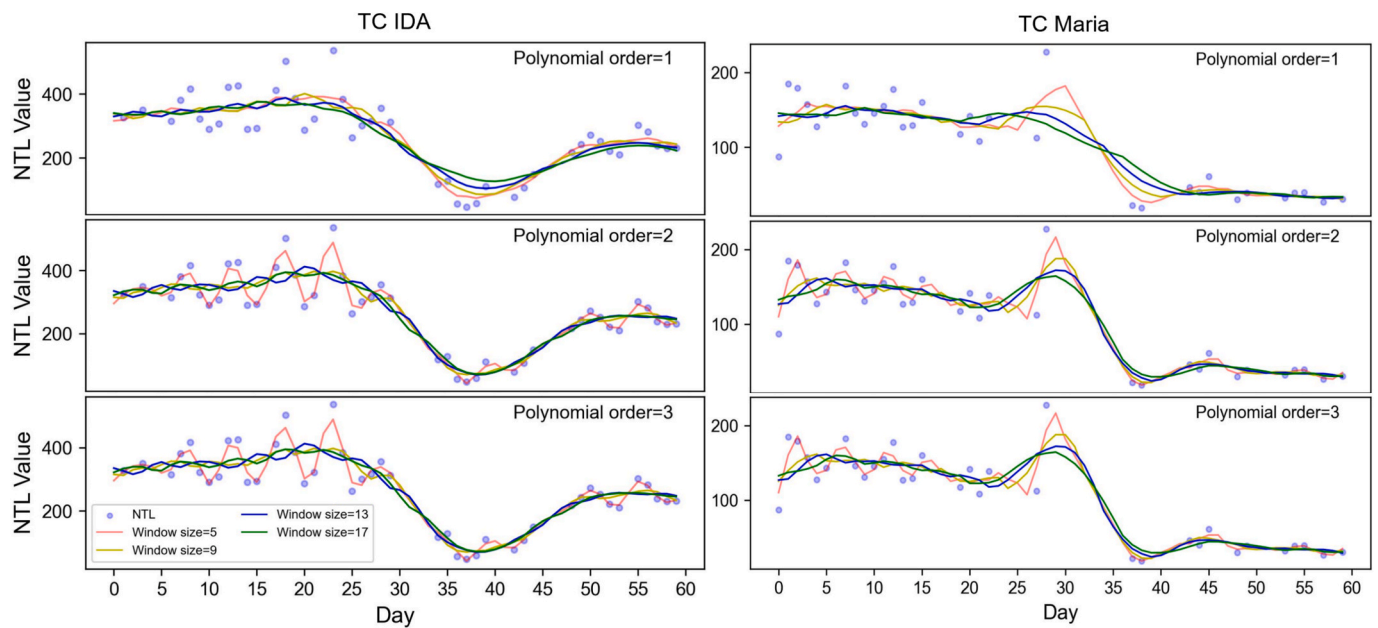


Fig. 5. Comparison of different Savitzky–Golay filter configurations applied to the NTL time series of two typical urban agglomerations. The optimal configuration (polynomial order = 2, window size = 9) effectively retains the minimum NTL values associated with TC impacts while suppressing high-frequency noise, offering a balance between signal fidelity and smoothing simplicity.

provided in the [Supplementary Materials](#) (Fig. S3).

3.3. Identification of agglomeration-TC events and TC impact onset

Urban agglomerations affected by TCs were identified using TC track points and their associated wind speed radii, referring to agglomeration-TC events hereafter. The IBTrACS dataset provides TC track data at 3-hour intervals, including wind speed radii corresponding to three wind speed thresholds (34 kt, 50 kt, and 64 kt) across four quadrants (northeast, southeast, southwest, northwest). To more accurately delineate the spatial footprint of each TC and ensure comprehensive identification of affected urban areas, the original 3-hourly track points and associated wind fields were interpolated to 10-minute intervals.

The 34 kt wind speed threshold is widely recognized as a standard metric for delineating the area affected by TC (Holland, 1980). For each interpolated track point, the average radius of 34 kt winds across the four quadrants (northeast, southeast, southwest, northwest) was calculated and used as the representative impact radius. A circular buffer was then constructed around each track point using this radius to

approximate the localized wind impact area. The cumulative TC impact zone was derived by aggregating the buffered areas of all interpolated track points (at 10-minute intervals), resulting in a high-resolution spatial footprint of each cyclone's influence. The time at which the first buffered zone intersected a given urban agglomeration was recorded as the TC impact onset time for that agglomeration (Fig. 6a).

3.4. Identification of agglomeration-PO events

In this study, agglomeration-PO events refer to POs in urban agglomerations identified via anomaly detection applied to NTL sequences (Fig. 6b). When a TC causes substantial damage to urban aggregation's power infrastructure, the corresponding NTL sequence typically exhibits a pronounced decline, reflected as a distinct low-value anomaly on the affected dates. Such abrupt drops are interpreted as statistical outliers, which may result from data input errors, extreme events, or natural variability (Barnett, 1994). Within this framework, agglomeration-PO events are treated as a subset of special-event anomalies and thus can be detected using statistical methods.

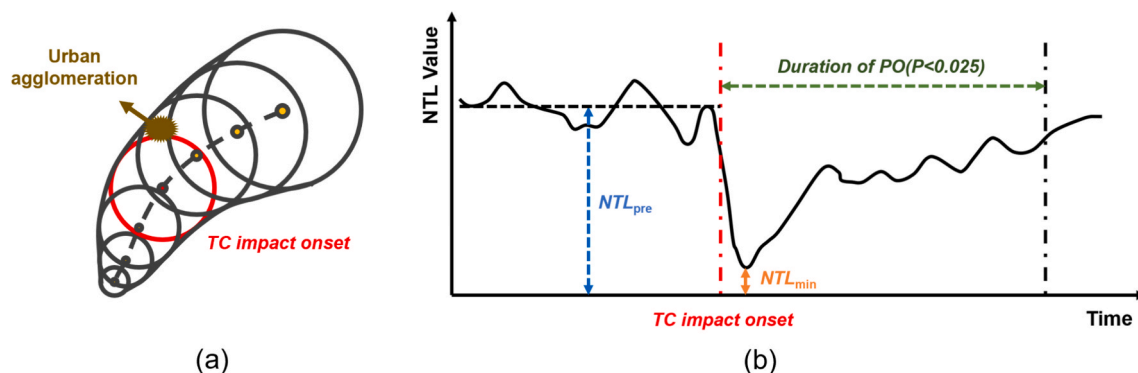


Fig. 6. (a) Determination of TC impact onset: the first interpolated track point whose 34 kt wind impact radius intersects an urban agglomeration is identified as the TC impact point (red), and its timestamp is defined as the TC impact onset. (b) Estimation of PO parameters: a probability density function is fitted to the two-month NTL sequence prior to TC impact onset to establish the pre-event baseline (NTL_{pre}). A cumulative probability threshold of 0.025 is used to detect PO events. The 180-day post-TC NTL sequence is analyzed to determine the PO duration, while PO severity is calculated as the relative loss in NTL, based on the minimum post-TC value (NTL_{min}) relative to NTL_{pre} .

Assuming a normal distribution, data points with cumulative probabilities below 0.025 or above 0.975 (equivalent to deviations exceeding ± 2 standard deviations from the mean) are generally considered statistical outliers (Barnett, 1994). Since typical NTL data in a given period generally follows a normal distribution, a lower-tail threshold of 0.025 was used in this study to identify agglomeration-PO events. To characterize typical NTL behavior under undisturbed conditions, a probability density function (PDF) was constructed using the NTL time series preceding the onset of each TC. This PDF served as the reference distribution against which post-TC deviations were assessed.

For reliable estimation of the distribution parameters, a minimum sample size of 30–50 observations is generally recommended when fitting a normal distribution (Bickel and Doksum, 2015; Wilcox, 2011). Accordingly, a two-month window of pre-TC NTL observations was selected to construct the PDF, balancing the need for sufficient data points while minimizing the influence of long-term variability. To further enhance the robustness of the baseline distribution, outliers within the pre-TC period were removed using an iterative filtering method (Cui et al., 2023). The resulting normal distribution of baseline NTL values is described as:

$$PDF = f(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad (2)$$

where μ and σ represent the mean and standard deviation of the pre-TC NTL sequence, respectively, and $f(x)$ is the fitted PDF of expected NTL values under normal conditions.

To estimate the duration of agglomeration-PO events, a 180-day post-TC NTL time series was extracted for each affected urban agglomeration. This extended observation window was selected to ensure full coverage of the recovery process and to accommodate the longest reported durations of urban power disruptions. To quantify deviations from baseline conditions, the cumulative probability P of each daily NTL value was calculated using the cumulative distribution function (CDF) derived from the pre-TC fitted PDF:

$$P = \int_{-\infty}^x f(x) dx \quad (3)$$

where x represents the observed daily NTL value, and $f(x)$ is the PDF fitted from the two-month NTL time series prior to TC impact onset (Eq. (2)). The probability P provides a statistical measure of how likely the observed NTL value is under normal, undisturbed conditions, thereby enabling the identification of abnormal low-light periods indicative of ongoing POs.

An agglomeration-PO event is identified when the cumulative probability P falls below the threshold of 0.025. The period during which P remains below this threshold is defined as the duration of PO, which reflects the core period of infrastructure disruption and recovery. In cases where no such anomaly was detected within the post-TC period, the TC was considered not to have caused PO, and the event is excluded from further analysis.

The PO severity was estimated as the relative loss in NTL value following the TC impact onset:

$$S = \frac{NTL_{\min} - NTL_{\text{pre}}}{NTL_{\text{pre}}} \quad (4)$$

where $NTL_{\min}$ is the minimum NTL value observed within the 180-day post-TC period, and NTL_{pre} is the mean NTL value calculated over the two months preceding the TC impact onset. The severity metric S quantifies the magnitude of the disruption, with higher values indicating more substantial power outages.

4. Results

4.1. Evaluation of the proposed method using the media reports

A total of 19,999 agglomeration–TC events were identified globally for the period 2012–2023 (Fig. 7a), representing instances where urban agglomerations were affected by tropical cyclones. Some agglomerations experienced multiple impacts from different TCs. Among these events, ~50% (10,022 events) cannot be evaluated for POs. This was mainly caused by the failure to obtain at least 30 valid observations within the two months preceding TC impact onset, which are required for constructing a statistically reliable PDF (Eq. (2)). For example, urban agglomerations in the Philippines frequently experienced persistent cloud cover and intense rainfall during the rainy season (typically May to November, Fig. 8a), driven by monsoonal and marine climatic influences. These conditions substantially reduced the availability of high-quality NTL observations, thereby preventing reliable PO detection. Although many sophisticated gap-filling methods have been proposed to recover or reconstruct nighttime light and other optical data (Hao et al., 2023; Yao et al., 2021; Ye et al., 2020), their validity in capturing abrupt events is unclear as such methods typically rely on assumptions of temporal continuity or spatial similarity. Among the remaining 9,977 events with sufficient valid NTL data, 1,239 large-scale agglomeration–PO events were identified, accounting for approximately 6% of all agglomeration–TC events. For these identified PO events, the onset of TC impacts, as well as the severity, and duration of POs were determined or estimated.

A total of 721 agglomeration–TC events reported by the media were compared with the algorithm's detection results (Fig. 7b). Among these, 136 events could not be evaluated due to insufficient NTL data. For the remaining 585 events, the confusion matrix (Fig. 7b) shows that 170 of the 201 media-reported outage events were correctly identified by the algorithm, while 290 of the 384 non-outage events were also accurately classified. Based on these results, the algorithm achieved a recall of 0.85, a precision of 0.64, and an overall accuracy of 0.79.

Most false negatives were concentrated in the Bohai Rim region of China and the Philippines. These omissions were largely due to missing NTL data during TC events, which impaired the algorithm's ability to accurately detect power outages—an issue clearly reflected in the spatial distribution of no-data occurrences in these areas. In addition, power outages in the Bohai Rim region are often restored rapidly, mostly within 1–2 days. This short disruption period, combined with data gaps, further contributed to omission errors. False positives generated by the algorithm were observed across various regions, with notable concentrations in East Asia, Europe, and the USA. Some of these errors were from the inherent limitations of the algorithm, particularly its tendency to misinterpret background noise as outage signals. This is more pronounced when identifying small-scale events. Moreover, false positives may also result from the uncertainty and incompleteness of media reports. Media coverage typically focuses on major or representative outages in key cities, while many smaller or peripheral agglomerations within the TC impact zone may be underreported or overlooked. This reporting bias complicates the validation process and may lead to apparent inconsistencies between algorithm outputs and media references.

True positive detections were most concentrated along the coastal regions of North America. This is likely due to the greater intensity of TCs in these areas, which often cause widespread and severe damage to power infrastructure. Outages in these regions tend to be longer in duration and broader in spatial extent, making them more easily detectable by the proposed method, even in the presence of cloud-contaminated data. In contrast, true negative detections were mainly distributed across northeastern South America, western Africa, western Europe, the coastal regions of the Persian Gulf, and southeastern Australia. In these regions, TCs tend to be weaker and have limited impact on power infrastructure, with power outages being infrequent or

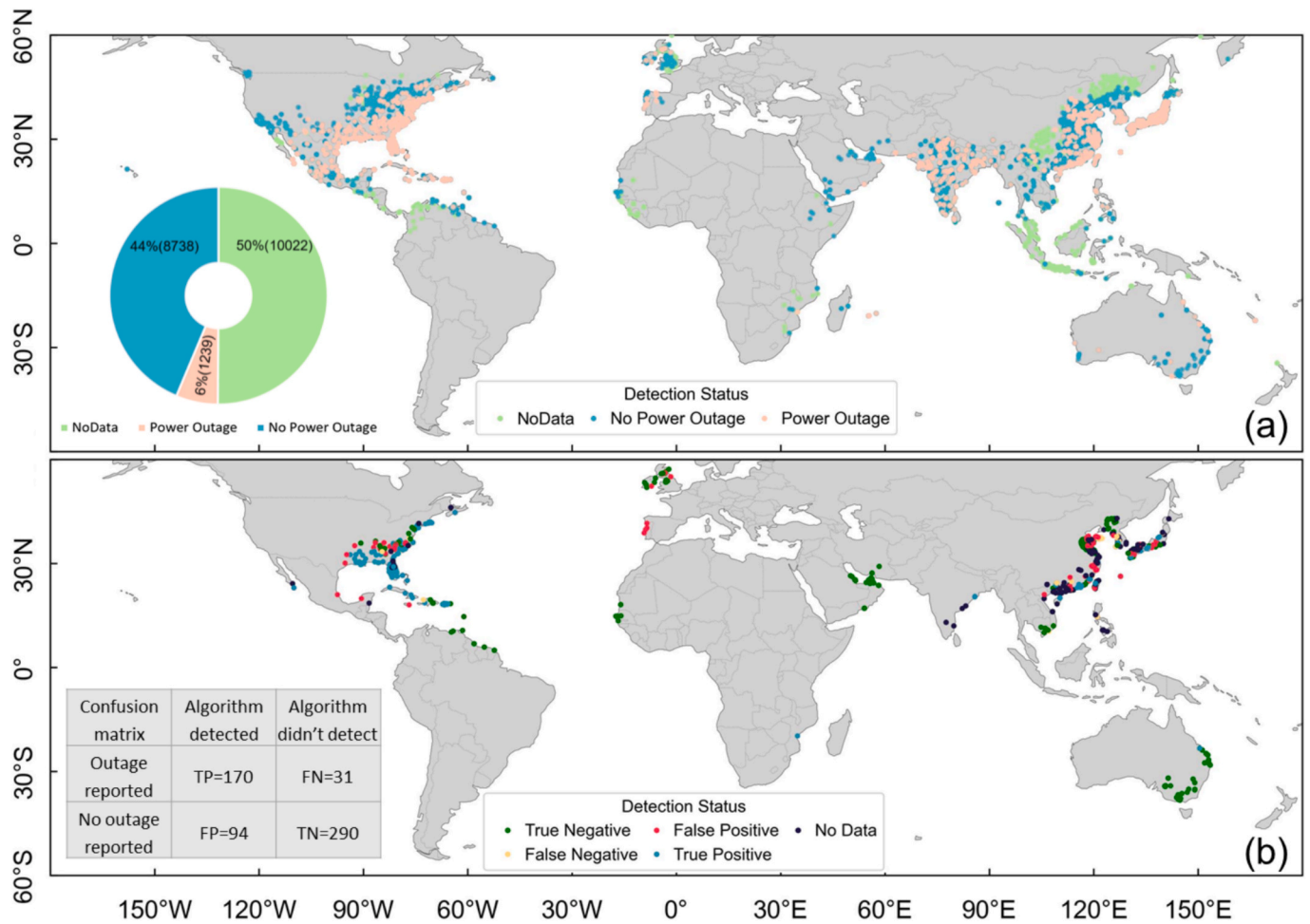


Fig. 7. (a) Spatial distribution of global agglomeration-PO events detected by the algorithm from 2012 to 2023. (b) Distribution of 721 agglomeration-TC events reported in the media with corresponding detection outcomes as True Negative, False Negative, False Positive, True Positive or No Data.

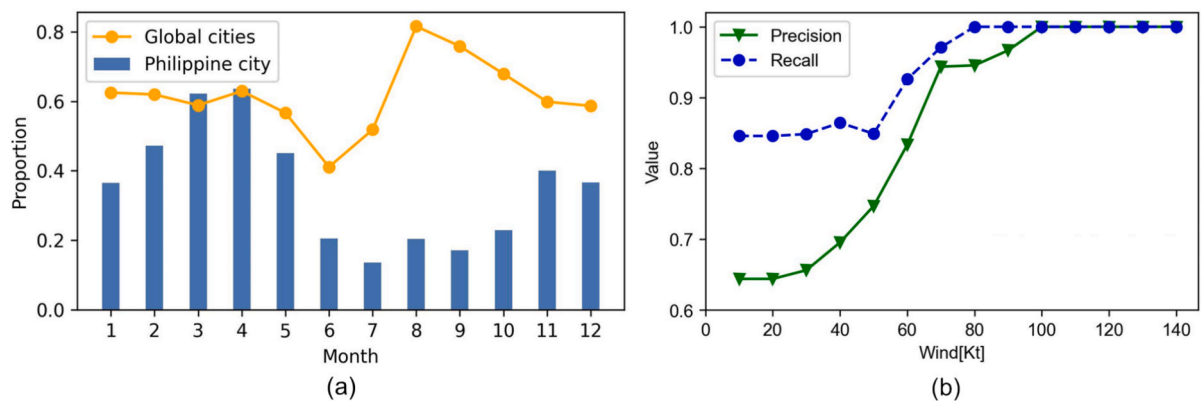


Fig. 8. (a) Monthly proportion of valid nighttime light (NTL) observations globally and for a selected city in the Philippines from 2012 to 2023, illustrating strong seasonal variability in data availability. (b) Performance of the proposed algorithm across different TC central wind speeds.

minor.

Fig. 8b shows how the recall and precision of PO detection vary with TC central wind speed. At lower wind speeds (<50 kt), the proposed algorithm achieved a recall rate of > 0.8, indicating that < 20% of outage events were not detected. In the same wind range, precision was comparatively lower (0.64–0.70), suggesting that approximately 30–35% of the detected outages were false positives. As aforementioned, they can be attributed to missing NTL data, data noise, the inherent

resilience of power infrastructure to low-intensity TCs, and the uncertainty or incompleteness of media reporting. The fact that recall consistently exceeded precision by around 0.2 implies that noise in the NTL signal and gaps in media reporting likely contribute more significantly to misclassification than do cloud contamination or rapid restoration alone.

A marked improvement in recall was observed when TC wind speeds exceeded 50 kt, indicating enhanced performance in detecting large-

scale POs associated with more intense storms. This trend aligns with empirical expectations, as higher-intensity TCs are more likely to cause extensive infrastructure damage, producing pronounced anomalies in NTL data that enhance the detectability of outages. Similarly, precision increased with wind speed, suggesting that at higher wind intensities, the impact of minor noise and the uncertainty of media reports on evaluation reduced. As wind speed further increased, recall reached 1 at 110 kt and precision reached 1 at 100kt, demonstrating the algorithm's strong capability to identify widespread outages under extreme wind conditions.

The difference between the estimated and reported TC impact onset times was calculated to assess the algorithm's temporal accuracy. The time differences ranged from -2 to $+2$ days, falling within an acceptable margin of error and indicating that the algorithm can effectively capture the onset of TC impacts. Since outage duration and severity were reported only for a limited number of typical TCs affecting major cities, the accuracy of the estimated durations and severities was further evaluated for these selected cases, as detailed below.

4.2. Analysis of agglomeration-PO events caused by typical TCs

A more detailed analysis was conducted on several representative

TCs to provide a comprehensive evaluation of the proposed method. Four well-detected cases and four poorly identified cases were examined. In addition, two additional case studies were included for inter-comparison (Mitsova et al., 2024; Román et al., 2019).

The four well-detected agglomeration-TC events occurred in regions typically characterized by inadequate power grid maintenance, vulnerable infrastructure, and limited post-disaster recovery capacity, all of which contribute to more severe and prolonged power disruptions (Fig. 9). In 2017, TC Maria made landfall in Puerto Rico with nearly Category 5 intensity (135 kt in the center), causing extensive damage to the local power infrastructure. Prior to the TC Maria, chronic deficiencies in power grid maintenance and systemic disorganization following utility restructuring had already made POs a recurring issue. The storm brought widespread flooding, landslides, and intense winds, destroying over 80% of the transmission infrastructure. This resulted in a prolonged blackout that lasted up to ten months, severely disrupting residents' livelihoods and delaying economic recovery (Kwasinski et al., 2019). The proposed algorithm estimated a power infrastructure damage level of 0.83 and a recovery period of 102 days for San Juan, the capital of Puerto Rico. This estimate is approximately 5% shorter than the 107 days recovery period reported by Román et al., which is based on the report of the grid system (Román et al., 2019). When compared

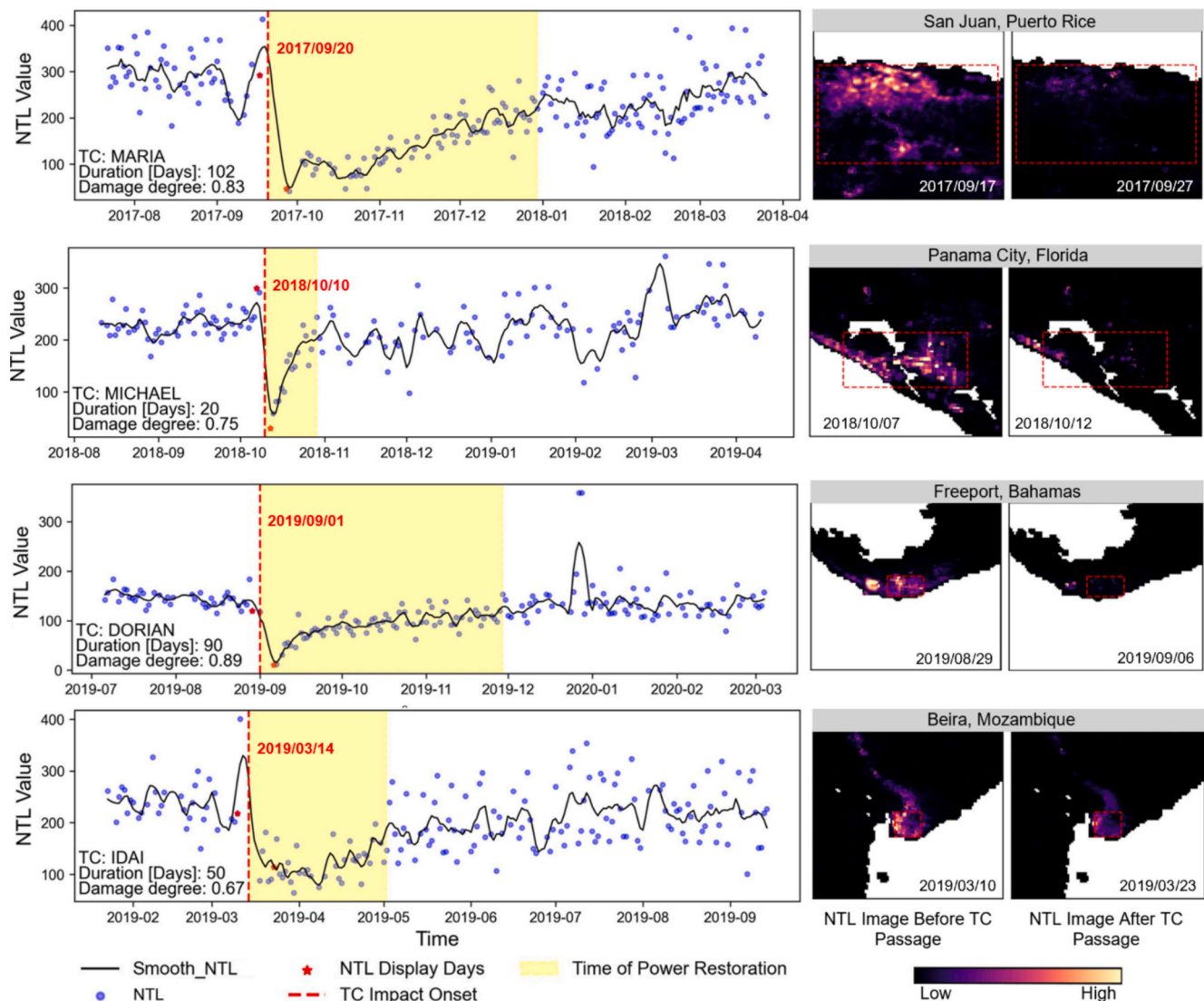


Fig. 9. NTL sequences of four typical agglomeration-TC events that were well captured by the proposed method, along with NTL images showing changes before and after the event.

with other small cities in Puerto Rico, the recovery durations estimated by the algorithm were approximately 5–35% shorter than those reported in the aforementioned study. For example, in Fajardo, the estimated duration was about 34% shorter, while in Humacao, the discrepancy was approximately 7%.

Another TC analyzed in this study is Hurricane Michael, which made landfall in October 2018, causing extensive damage across Florida and resulting in widespread POs (Beven et al., 2019). The proposed algorithm identified a prolonged outage lasting 20 days in Panama City. A notable spatial agreement was observed between the affected areas identified in this study and those reported by Mitsova et al., (Mitsova et al., 2024). Comparative analysis indicated that in Bay City, the algorithm estimated an outage duration of 23 days, approximately 13% longer than the 20 days reported by Mitsova et al., In contrast, the algorithm largely underestimated the outage duration in Leon City, detecting only 4 days of outage compared to the 8 days reported in the reference study. The proposed algorithm generally produced shorter outage duration estimates compared to those reported in ground-survey-based studies, primarily due to differences in the definitions of outage duration. In this study, outage duration is defined as the period during which NTL anomalies indicate disruptions to the power system, rather than the entire period required for full restoration. As a result, the

algorithm captured the main disruption phase rather than the complete recovery process. Additional discrepancies may also arise from differences in spatial aggregation approaches. This study delineates affected areas based on urban agglomerations, whereas previous studies often use county-level administrative boundaries. Despite these methodological differences, the relative distribution of outage durations across cities remained broadly consistent with previous findings.

Two additional TCs with good agreement between the algorithm estimates and reported data were TC Dorian and Idai, both in 2019. TC Dorian struck Freeport, Bahamas with maximum sustained winds of 161 kt and caused severe damage and major POs. In some areas, the destruction was so severe that restoring power was very difficult (Zegarra et al., 2020). The proposed algorithm produced estimates that align closely with these observations, indicating a power infrastructure damage level of 0.89 and a recovery period of 90 days for Freeport. TC Idai made landfall in Mozambique, Africa in March 2019 with wind speeds of ~ 105 kt. It caused severe flooding with water levels above three meters. The floodwaters took several weeks to recede, leading to extensive infrastructure damage and a cholera outbreak (Comino 2021). Damage from TC Idai to power infrastructure tends to be more severe than TCs with similar intensities in other regions due to the vulnerability of the grids and limited emergency response capacity in Mozambique.

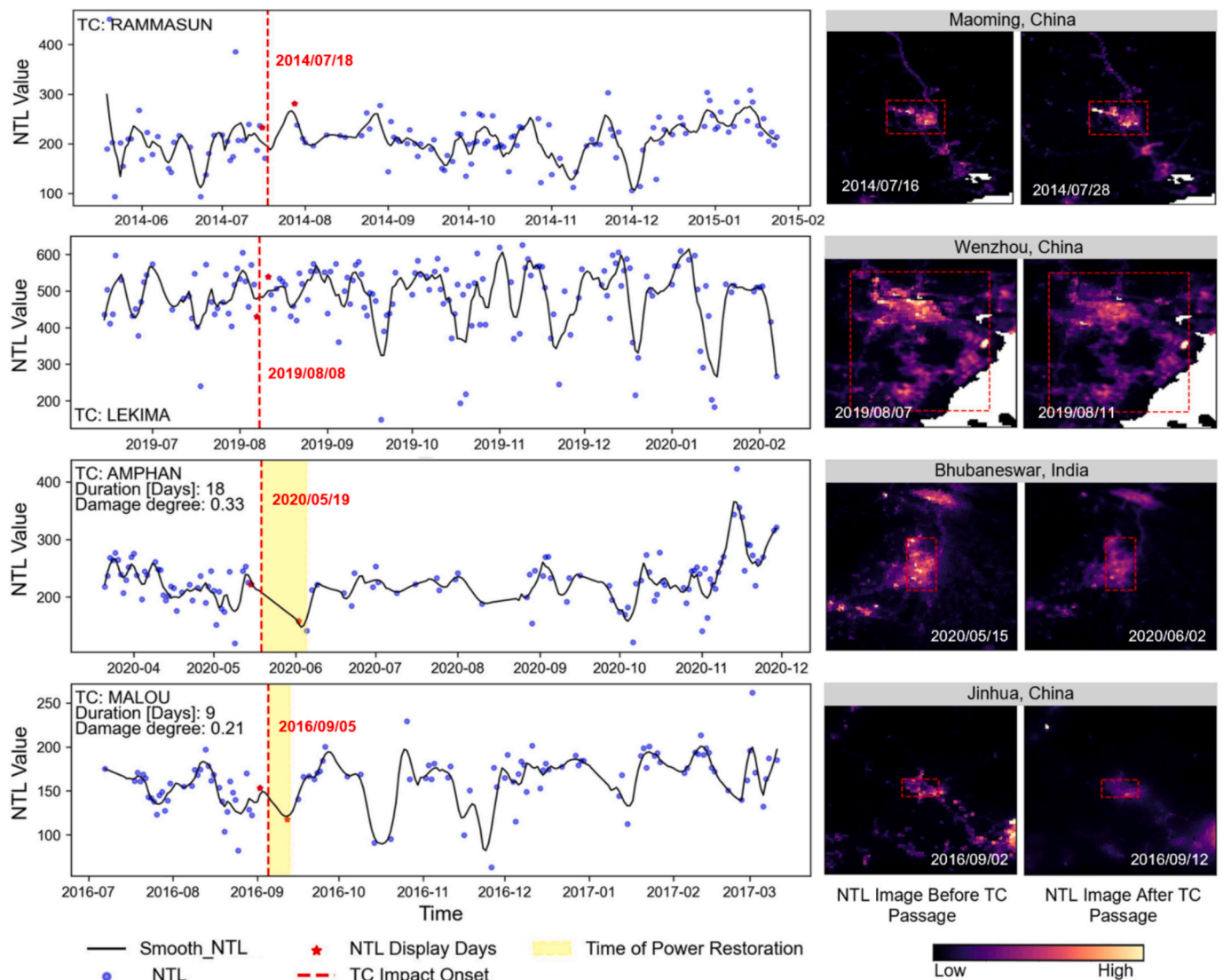


Fig. 10. NTL sequences of four typical agglomeration-TC events that were poorly captured by the proposed method, along with NTL images showing changes before and after the event.

The algorithm estimated power infrastructure damage levels of 0.67–0.86 for the affected urban agglomerations, with recovery periods of 42–50 days.

In regions with strong disaster resilience, PO patterns tend to more complex, often involving automatic pre-emptive shutdowns and rapid repair efforts. These factors make it challenging for accurately estimating the severity and duration of outages from NTL (Fig. 10). For

example, although Super Typhoon Rammasun severely damaged the power grid in southern China in 2014 (Yuan et al., 2020). The prompt response by the China Southern Power Grid limited the drop in NTL brightness, making detection difficult. A similar situation occurred in Wenzhou, China during super TC Lekima in 2019. Despite widespread outages being reported (Zhou et al., 2022), the rapid restoration of transmission lines and substations resulted in minimal detectable

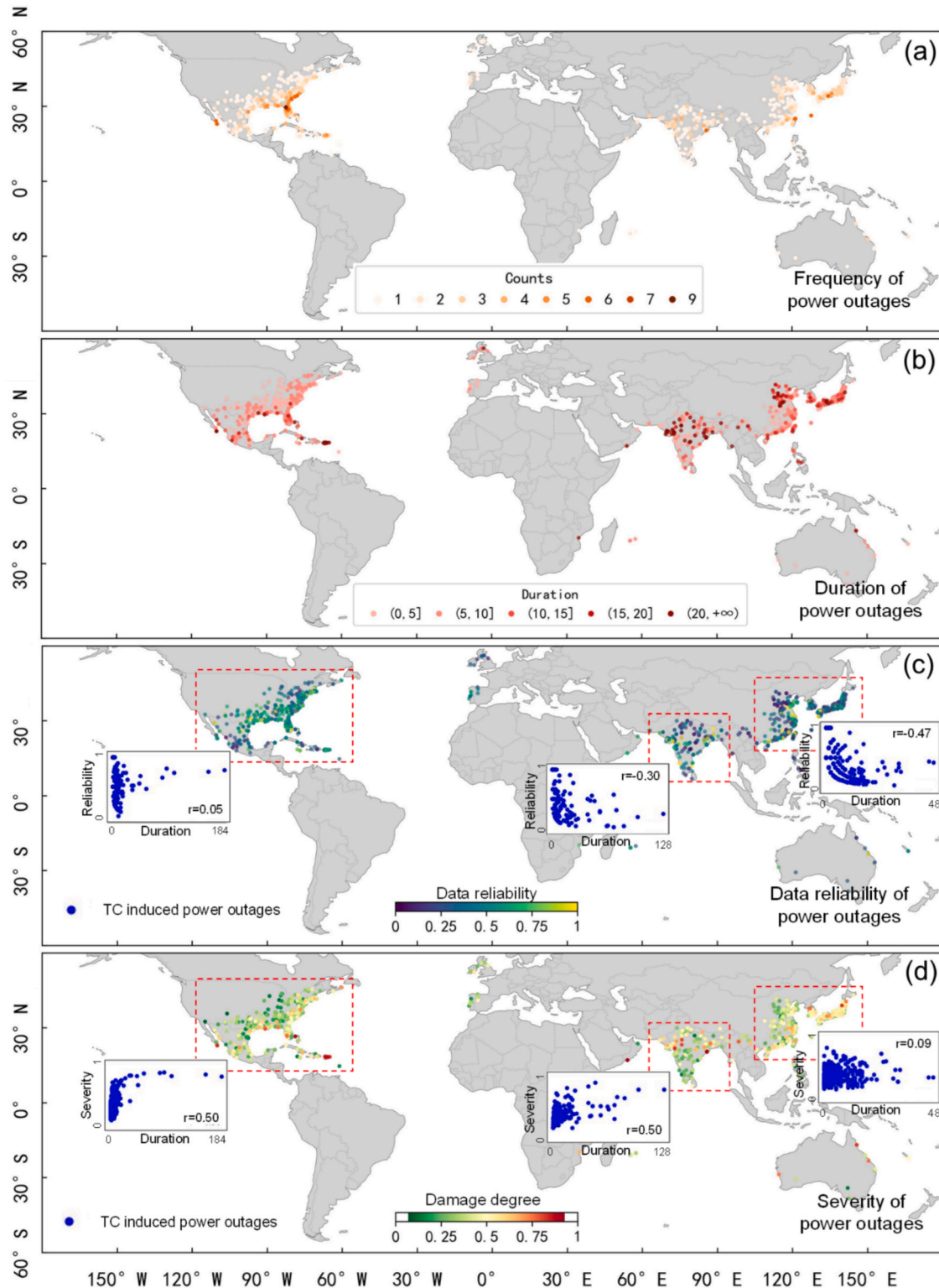


Fig. 11. TC-induced POs across global urban agglomerations from 2012 to 2023. (a) to (d) are the frequency, duration, data reliability and severity of POs respectively. Regional patterns of the relationships between duration and both data reliability and severity are also shown for North America, India, and East Asia.

changes in NTL. The proposed algorithm also struggled in areas with poor data quality. For example, during TC Amphan in 2020, extended data gaps in Bhubaneswar, India, led to underestimation of outage severity, as power was restored before NTL observations resumed. In other cases, the algorithm misclassified noise as outages. One such instance was during TC Malou's impact on Jinhua, China, in 2016, where minor damage occurred, but poor data quality and missing values led to false detection of an outage.

4.3. Spatial patterns of the identified agglomeration-TC events

The spatial distribution of agglomeration-TC events from 2012 to 2023 showed clear regional differences (Fig. 11). East Asia, India, and the eastern and southern parts of North America experienced the highest outage frequencies (Fig. 11a). Frequent TC exposure combined with dense urban populations increased the risk of power disruptions in these regions. In East Asia, most affected cities experienced < 5 outages over 12 years, though a few areas recorded 6 to 7 events. A similar pattern was seen in India. In contrast, the southeastern United States showed much higher outage frequencies. This vulnerability results from frequent intense TCs, dense populations, and concentrated infrastructure, leading to longer and more severe outages. Some urban agglomerations experienced up to 9 outages, with > 5 outage events being relatively common. These findings highlight that the southeastern United States faces not only high TC risk but also more prolonged and severe power system disruptions.

The spatial distribution of outage durations reveals (Fig. 11b) that agglomeration-PO events lasting more than 15 days are mainly concentrated along the southeastern coast of North America. In contrast, in India and East Asia, such events are primarily located in inland regions. This counterintuitive pattern highlights the need to further assess the reliability of the algorithm's results. Since the accuracy of outage detection depends heavily on the completeness and quality of NTL data, the proportion of valid observations during each detected outage period was calculated as an indicator of data reliability (Fig. 11c). This metric reflects how complete and interference-free the remote sensing data are during critical periods. Higher values indicate greater confidence that the detected outage is real rather than a result of data noise. The southeastern coast of North America demonstrated relatively high data reliability, with most events showing values above 0.75. This indicates the detected long-duration outages were well supported by high-quality remote sensing data and strong signal-to-noise ratios in the region. In contrast, reliability values in India and East Asia were generally lower, especially for events lasting more than 15 days, many of which had values below 0.25. Some of these long-duration outages may be false positives caused by data gaps or noise. In India and East Asia, outages shorter than 15 days had relatively higher reliability scores of > 0.5, being reliable for further applications.

A joint analysis of outage duration and severity also revealed notable interregional differences (Fig. 11d). TC-induced POs in North America and India were generally more severe than those in other regions, with a higher proportion of events exhibiting severity values of > 0.75. Here, "severity" is defined as the relative reduction in NTL brightness compared to baseline levels, used as a proxy for the extent of damage to power infrastructure. A positive correlation was observed between outage severity and duration in this region, indicating that more severe outages often lasted longer. This pattern reflects the widespread and intense damage to transmission lines, substations, and distribution networks caused by major TCs. In contrast, outages in East Asia were typically less severe, with most severity values below 0.75. The correlation between duration and severity was also weaker and less consistent. In several cases, outages lasted more than 15 days despite having low severity—an uncommon pattern, as prolonged outages usually correspond to more extensive damage. When examined alongside data reliability scores, these anomalies were often linked to poor observation quality, suggesting that such inconsistencies likely stem from limitations

in the NTL dataset rather than actual power system performance.

5. Discussion

This study developed a globally applicable PO detection algorithm based on NTL data, enabling the systematic mapping of TC-induced outages. By utilizing the daily-updated NASA Black Marble product (Wang et al., 2024), the algorithm can support near-real-time monitoring and significantly reduces reliance on manual data collection, which often delays conventional damage assessments (Zheng et al., 2025). Unlike previous work that focused primarily on the United States (Zheng et al., 2025) or isolated case studies (Mitsova et al., 2024; Román et al., 2019), this study provided the first global-scale, consistent analysis of TC-induced urban power disruptions. Although the present method is not designed to support the detailed analysis of intra-county inequality and pixel-level spatial heterogeneity following hurricane events (Zheng et al., 2025), it allows for the automated estimation of outage onset, duration, and severity across diverse geographies and meteorological conditions. This enables a broader assessment of how disaster impacts vary between different countries and urban systems, providing a standardized global perspective that complements localized, fine-scale recovery studies.

This study presents the most comprehensive evaluation to date of TC induced PO detection using media reports. Validation was conducted across 721 city–typhoon (TC) events involving 88 TCs, demonstrating that the proposed algorithm achieved an overall precision of 0.64 and recall of 0.85. Both metrics exhibit clear improvement with increasing TC wind speed, reflecting better detectability of more severe outages. These findings underscore the algorithm's effectiveness in identifying major disruptions, particularly in urban agglomerations frequently impacted by intense cyclones. However, comparisons with existing studies (Mitsova et al., 2024; Román et al., 2019) and media reports suggested that the algorithm tends to underestimate outage durations (mostly < 15%). This discrepancy is primarily from the different definitions of outage durations, with the proposed method focusing on the core period of disruption rather than the full duration of the recovery process. Because stray and small-scale power shortages cannot be captured by the NTL.

Case studies of representative TC events across different continents further demonstrated both the strengths and limitations of the method. In regions such as the southeastern United States and parts of the Caribbean (Fig. 9), where TCs frequently cause widespread physical damage to power infrastructure, the algorithm performed well in capturing both the spatial extent and temporal dynamics of outages. For example, long-duration outages in Puerto Rico and Florida were well detected, with estimated recovery periods aligning closely with documented records. In these cases, infrastructure vulnerability, limited maintenance, and delayed restoration contributed to extended blackout durations, enhancing the algorithm's sensitivity to disruption.

In contrast, performance was reduced in regions characterized by rapid power restoration, low-intensity TCs, or poor data quality (Fig. 10). In parts of India, East Asia, and Southeast Asia, outages of more than 15 days were occasionally detected despite low severity levels—likely reflecting false positives due to missing or low-quality observations rather than actual damage. Persistent cloud cover and frequent NTL data loss during TC events further compromised detection reliability in these areas. While many gap-filling methods are effective for gradual or predictable data gaps (Hao et al., 2023; Liu et al., 2023; Yao et al., 2021), e.g., interpolation, machine learning-based reconstruction, or spatiotemporal smoothing, they often struggle with abrupt, localized events. Rather than applying a temporal gap-filling method, TC-city events with insufficient valid observations were excluded from the analysis. For events with "adequate" data coverage, data reliability scores were calculated based on the proportion of valid observations during the outage period. These scores revealed that the accuracy of the detection algorithm declined significantly when data availability

dropped below 0.5. This finding highlights the need for careful interpretation in regions with limited data quality or rapid post-disaster recovery, where nighttime light signals may not reliably reflect outage conditions.

The use of a 9-day moving average window helped reduce short-term noise and enhance robustness but introduced temporal smoothing, making it difficult to pinpoint the exact timing of power restoration. While the algorithm was able to estimate outage onset within ± 2 days of reported landfall impacts in most cases, this temporal uncertainty limits its utility for precise event timing. Furthermore, the spatial resolution and atmospheric sensitivity of the NTL product imposed inherent constraints. Factors such as moonlight, aerosols, and cloud cover can introduce artifacts or obscure real signals, resulting in either missed detections or false positives (Montoya-Rincon et al., 2022). Recent studies have also noted NTL data's limited sensitivity to changes in dimly lit rural areas (Bara and Sticher, 2025), yet this constraint is negligible here given our focus on high-radiance urban core zones delineated from land-surface datasets.

A primary source of uncertainty in the validation of global PO events arises from the spatial mismatches between urban agglomerations and the administrative units cited in media reports. In this study, the 19,999 identified agglomeration–TC events were delineated by the spatial intersection of an agglomeration's centroid with the 34-kt wind radius of a tropical cyclone (Fig. 6a). However, because media reports were organized by administrative boundaries, such as cities, provinces, or states, a precise one-to-one mapping is not always feasible. The uncertainties associated with such spatial mismatches were analyzed through different scenarios (Table 1). Among the 201 media-reported outage events, 91 explicitly identified affected cities, whereas the remaining 110 reports described impacts only at the provincial or state level. For these state-level reports of TC-induced POs, all agglomerations within the affected administrative unit were treated as positive cases. However, it is probable that not all agglomerations within a given state experienced outages in reality.

Regarding precision, the metric is likely underestimated in this study. If the model correctly identifies an agglomeration as stable, but that specific unit was labeled as an affected zone by a broad state-level report, the result is statistically recorded as a false positive. Because the denominator for precision includes false positives, this mismatch makes the algorithm appear less precise than its actual performance. The impact on recall is similarly influenced by the interaction between the predictions and the falsely labeled positives. If the algorithm were to identify outages for these stable cities, the recall would be artificially inflated. However, because the proposed algorithm is designed to detect significant Nighttime Light (NTL) deviations, it is more likely that the algorithm will correctly identify these stable cities as negatives. In this scenario, the model is penalized with a false negative for being more accurate than the generalized state-level report. Consequently, the total number of expected outages in the denominator is artificially increased,

leading to an underestimation of recall.

Despite these potential underestimations of performance, the reported metrics offer a conservative and reliable measure of the algorithm. In the context of disaster monitoring, maintaining a high recall is of particular importance because it reflects the proportion of true events successfully identified. The failure to detect a true outage is generally considered more consequential than the over-identification of potential impact zones. Therefore, the metrics presented here represent a lower-bound estimate of the true capability of the model, ensuring that the conclusions regarding global power recovery patterns are supported by a rigorous validation framework.

Similar uncertainties exist in the validation of non-outage events (Table 1). Among the 384 media-reported non-outage cases, 115 involved weak or unnamed cyclones that were unlikely to cause city-scale infrastructure failure. These were utilized as precise validation data, and the evaluation of these cases was considered highly reliable. The remaining 269 cases involved moderate-to-strong cyclones for which local outages may have occurred but remained unreported in media accounts. In such scenarios, if the proposed method identifies a power outage in an area lacking corresponding media coverage, the detection is statistically categorized as a false positive. This leads to a potential underestimation of precision, as the model may be detecting actual localized outages that the media failed to record.

Conversely, this incomplete reporting could theoretically lead to an inflation of recall estimates, as the pool of expected outages in the validation set is smaller than the actual number of events on the ground. Nevertheless, these unreported outages were generally minor and unlikely to reach the threshold for city-scale or state-scale reporting. As a result, their influence on the overall evaluation is limited, and the assessment remains robust. The metrics therefore provide a conservative yet reliable measure of performance, ensuring that the principal conclusions regarding global power recovery patterns are unaffected by these minor reporting uncertainties.

Future studies may enhance validation by incorporating information from social media platforms, which can offer more timely, localized, and granular accounts of power outages, especially in regions with limited media coverage. Extending the historical analysis using DMSP-OLS data may also be feasible for earlier TC events (1992–2012), though additional challenges require to be addressed due to the product's lower spatial and radiometric resolution. Combining NTL data with infrastructure maps, restoration reports may further improve detection reliability and reduce temporal ambiguity. Moreover, the current method is grounded in statistical theory and relies on threshold-based detection, which makes it more sensitive to missing data and noise. Deep learning approaches may offer greater robustness by learning complex patterns and compensating for data gaps or observational uncertainty.

6. Conclusion

This study developed a global-scale PO detection algorithm using satellite-based NTL data, enabling the automated identification of urban outages triggered by TCs. By applying the algorithm to NASA's Black Marble product from 2012 to 2023, a total of 1,239 TC-induced PO events were detected, along with estimates of their onset time, severity, and duration. Validation using 721 media-reported agglomeration–TC events shows that the algorithm achieved a recall of 0.85, precision of 0.64, and overall accuracy of 0.79. Detection performance improved with increasing TC wind speed, reflecting the algorithm's capability to reliably identify large-scale, high-severity outages. Case studies across regions such as the southeastern United States and Puerto Rico revealed strong consistency with historical outage records, particularly in areas with vulnerable infrastructure and high-quality NTL observations.

However, the algorithm's performance is constrained in regions with frequent data gaps, rapid restoration, or low-impact events, where outage signals are less detectable. Despite these challenges, the proposed approach offers a cost-effective, scalable, and near-real-time method for

Table 1
Impact of spatial and reporting uncertainties on validation metrics.

Scenario	Details	Impact on precision	Impact on recall
City-level PO reports	Direct matching between agglomerations and explicitly reported cities.	Marginal impact	Marginal impact
State-level PO reports	Broad matching where all agglomerations in a state are labeled as affected.	Underestimated	Underestimated
Unreported POs (weak TCs)	Real outages occurring during weak storms that were missed by media.	Marginal impact	Marginal impact
Unreported POs (strong TCs)	Real outages occurring during major storms that were missed by media.	Underestimated	Potentially inflated

global TC-induced outage monitoring. It provides valuable insights for assessing power system resilience and supporting disaster preparedness and recovery planning.

CRedit authorship contribution statement

Liujun Zhu: Writing – review & editing, Writing – original draft, Conceptualization. **Yaqian Li:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation. **Shanshui Yuan:** Writing – review & editing, Supervision, Resources, Conceptualization. **Shi Shi:** Writing – review & editing, Formal analysis, Data curation. **Fang Ji:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The media records of TC-induced POs and the effect of using different S-G filters on the PO identification were provided in Supplementary materials. Supplementary data to this article can be found online at <https://doi.org/10.1016/j.isprsjrs.2026.01.042>.

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